Generalized Bayes: the route back to standard Bayes

The inference engine FedGVI federates is the generalized (Gibbs) posterior (Bissiri et al. 2016; Jiang and Tanner 2008; Jewson et al. 2018; Knoblauch et al. 2022),

Generalized Bayes (part 2)

which trades the likelihood for a loss L and the KL regularizer for a general divergence $\mathcal{D}$:

Generalized Bayes (part 3)

$$q_n^*(s) = \arg\min_{q_n} \left\{ \begin{array}{c} \mathbb{E}_{q_n}\Big[\sum_i L(s; o_i) \Big] \\ + \frac{1}{\tau}\, \mathcal{D}\big(q_n \,\|\, \pi_0\big) \end{array} \right\}. \qquad (2)$$

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Generalized Bayes (part 4)

with prior π_0, learning rate τ, and regularizing divergence $\mathcal{D}$. The learning rate is part of the inferential specification, not a cosmetic constant;

Generalized Bayes (part 5)

coarsened-posterior and safe-Bayes work show why calibration of that temperature matters under misspecification (Miller and Dunson 2018; Grünwald 2012), where ordinary Bayes concentrates around a KL pseudo-truth rather than literal truth when the model family is wrong (Kleijn and Vaart 2012).

Generalized Bayes (part 6)

We name the object eq. 2 defines.

Generalized Bayes (part 7)

Definition (Generalized-(Gibbs)-Bayes posterior)

Given loss L, prior π_0, learning rate $\tau > 0$, and divergence $\mathcal{D}$, the generalized-Bayes posterior q_n^* minimizes (2). When $\mathcal{D} = \mathrm{KL}$, its exact categorical minimizer is the tempered softmax in (3).

Generalized Bayes (part 8)

$$q_n^*(s) \;\propto\; \pi_0(s)\,\exp\!\big(-\tau \textstyle\sum_i L(s;o_i)\big), \qquad (3)$$

Generalized Bayes (part 9)

The implementation surface is the `generalized_posterior` function in the `generalized_bayes` module.

Generalized Bayes (part 10)

The tempered softmax of eq. 3, stated in the definition above, is not an approximation: it is the exact closed-form minimizer of eq. 2 when the regularizer is the KL divergence, because the categorical support is finite and the objective is strictly convex in q.

Generalized Bayes (part 11)

The recovery to standard Bayes follows by choosing the loss.

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Generalized Bayes (part 12)

With $L = \mathrm{NLL}$, $\mathrm{NLL}(p, o) = -\log p(o)$, the exponential in that tempered softmax becomes a product of likelihoods and the minimizer is *exactly* standard Bayes; eq. 11 in sec. 8.8 states that corner,

Generalized Bayes (part 13)

and Corollary 6 there pins it to the closed-form prior-times-likelihood product.

Generalized Bayes (part 14)

The largest observed discrepancy between `generalized_posterior` in this regime and the analytic Bayes posterior is 5.55e-17, reported in sec. 10.1 — exact to machine precision (a maximum deviation of about one ULP), not merely close.

Generalized Bayes (part 15)

FedGVI computes each client update against a *cavity* rather than the full posterior, so a contributing agent does not double-count its own previous message. We name that operation.

Generalized Bayes (part 16)

Definition (Cavity / PVI factor update)

The cavity is the normalized removal of agent n's site factor from the colony posterior in natural-parameter space, as in (4). PVI replaces the site factor and recombines it with that cavity according to (5).

Generalized Bayes (part 17)

$$q_{-n}(s) = \frac{q(s)/t_n(s)}{\sum_{s'} q(s')/t_n(s')} = \text{softmax}(\log q(s) - \log t_n(s)), \tag{4}$$

Generalized Bayes (part 18)

The final expression makes normalization explicit. Taking a cavity and re-multiplying the original site factor restores the global posterior:

Generalized Bayes (part 19)

$$q(s) = \frac{q_{-n}(s)t_n(s)}{\sum_{s'} q_{-n}(s')t_n(s')}, \tag{5}$$

Generalized Bayes (part 20)

This is the recombination identity satisfied by `generalized_bayes.cavity` and `generalized_bayes.update_factor`.

The numbered recombination identity is eq. 5.

Generalized Bayes (part 21)

The cavity of eq. 4 is the discrete analogue of the expectation- propagation / partitioned-VI cavity used outside active inference (Ashman et al. 2022; Bui et al. 2018),

Generalized Bayes (part 22)

imported here so that the per-agent generalized-Bayes update of eq. 2 is computed against the colony belief with the agent's own contribution removed — exactly the sensory-attenuation discipline the belief-sharing round of sec. 8.9 requires.

Generalized Bayes (part 23)

What remains unspecified in eq. 2 are its two ingredients — the divergence $\mathcal{D}$ and the loss L — whose robust members and standard-Bayes limits sec. 8.6 develops next;

Generalized Bayes (part 24)

the aggregation identity (sec. 8.8) then federates the resulting per-agent posteriors.

Generalized Bayes (part 25)

The authoritative notation supplement makes the same normalization and recombination contract explicit in eq. 36 and eq. 37; those equations govern the symbols used by the implementation and all later supplements.

Conjugate likelihood learning for the shared model

Active-inference agents learn the parameters of their generative model, not just plan with them (Smith et al. 2022; Friston et al. 2024).

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Conjugate likelihood... (part 2)

The likelihood matrix A carries a Dirichlet prior with concentration a over each column, updated conjugately by accumulating observation-state co-occurrence counts (eq. 18),

Conjugate likelihood… (part 3)

giving the column-normalized expected likelihood. The update of eq. 18 is driven by the expected sufficient statistics under the data-generating model,

so as the concentrations accumulate $\mathbb{E}[A]$ converges to the true likelihood.

Conjugate likelihood... (part 4)

Convergence is measured by the per-column KL divergence summed over hidden states, which decreases monotonically toward the standard-Bayes fixed point; sec. 10.3 reports the learning curve, where the KL falls from 3.4231 to 0.0027 across 24 count batches.

Conjugate likelihood... (part 5)

A forgetting hyperprior optionally decays the running mass toward an asymptote so the agent does not become infinitely confident; with the hyperprior disabled the classical unbounded accumulation of eq. 18 is recovered.

Conjugate likelihood... (part 6)

The implementation is `learn_likelihood` in the `dirichlet_learning` module.

Bayesian model reduction for structure comparison

Structure learning in the active-inference frame proceeds by Bayesian model reduction (BMR): given a full model with Dirichlet posterior `post` under prior `prior`,

Bayesian model reduction… (part 2)

the change in negative variational free energy from swapping in a *reduced* prior — for example one that prunes a redundant column toward zero —

Bayesian model reduction... (part 3)

is available in closed form without re-running inference (Friston and Penny 2011; Smith et al. 2022).

Bayesian model reduction... (part 4)

Because the likelihood is shared, the reduced posterior is `post + reduced_prior - prior`, and the free-energy difference is a difference of log multivariate Beta functions (eq. 20), where $\ln B(a) = \sum_k \ln \Gamma(a_k) - \ln \Gamma(\sum_k a_k)$ is the log Dirichlet normalizer.

Bayesian model reduction... (part 5)

A positive ΔF in eq. 20 means the reduced model has more evidence — the pruned structure was redundant and should be adopted; a negative ΔF means the reduction destroyed something the data support. When the reduced prior equals the prior the score is identically zero,

Bayesian model reduction... (part 6)

the no-reduction fixed point.

sec. 10.4 reports $\Delta F = 3.68$ for a redundant reduction (accepted) against $\Delta F = -27.67$ for a supported one (rejected). The implementation is `bayesian_model_reduction.reduce`.